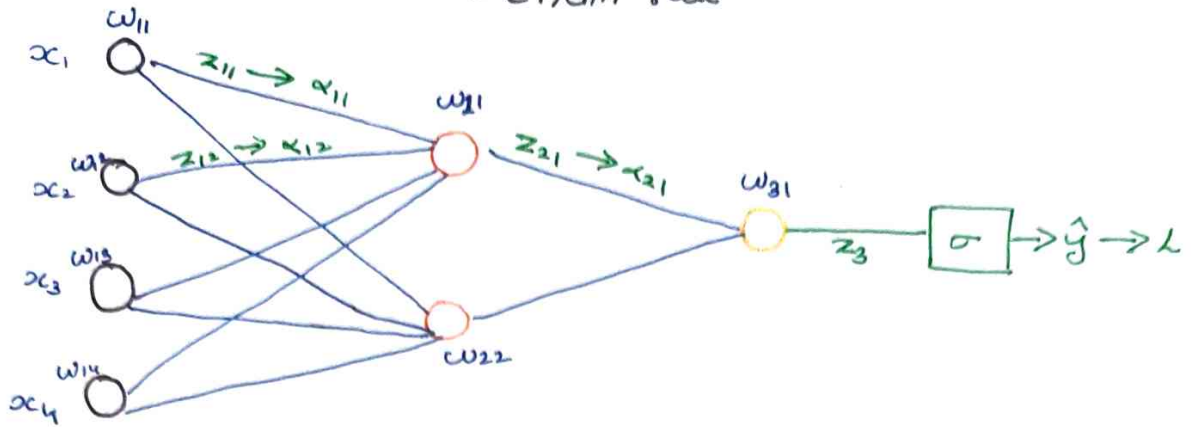


CNN Back propagation - Chain rule



$$\frac{\partial L}{\partial x_1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_3} \times \frac{\partial z_3}{\partial \alpha_{21}} \times \frac{\partial \alpha_{21}}{\partial z_{21}} \times \frac{\partial z_{21}}{\partial \alpha_{11}} \times \frac{\partial \alpha_{11}}{\partial z_{11}} \times \frac{\partial z_{11}}{\partial x_1}$$

$$\frac{\partial L}{\partial x_2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_3} \times \frac{\partial z_3}{\partial \alpha_{21}} \times \frac{\partial \alpha_{21}}{\partial z_{21}} \times \frac{\partial z_{21}}{\partial \alpha_{12}} \times \frac{\partial \alpha_{12}}{\partial z_{12}} \times \frac{\partial z_{12}}{\partial x_2}$$

⋮

$$\frac{\partial L}{\partial x_n}$$

$n \rightarrow$ number of flattened values from CNN

↓ Relu derivation $x + ve \rightarrow 1$ $x - ve = 0$

Convert to matrix
multiply with pixel it impacts and sum the
change filter weights.